

Olympic Success Around the World

Narrative Document

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1. Description of the Data Story

This project investigates how the success of Olympic medalist's changes over time and between countries. The analysis examines the correlation between economic resources, geography, and Olympic performance, utilizing two datasets from Kaggle: An Olympic Games athlete events dataset (`olympic_games.csv`) and a GDP per capita dataset (`gdp-per-capita-worldbank.csv`). The main question is whether richer countries always win more medals, and how the way the Summer and Winter Games are set up affects those results.

There are four visualizations that tell the story: a bar chart showing the number of medals won by each country, a scatter plot comparing GDP per capita to the number of medals won, a geographic medal map, and a line chart showing how medal records have changed over time by season. These charts show that Olympic success is not evenly spread around the world it is mostly found in rich, densely populated countries with established sports infrastructure. The Summer Games are much bigger than the Winter Games, which has a big effect on it.

2. Descriptions of Each Chart

Chart 1: Medal Count by Country

This information comes from `olympic_games.csv` (Kaggle). This stacked bar chart shows how many bronze, gold, and silver medals a group of the best-performing countries has won. The United States is very different from the other countries in the comparison because it has a lot

more medals. Australia, Canada, France, Germany, Great Britain, Hungary, Italy, Japan, Sweden, and the United States are also doing well.

This chart shows that only a few countries have a lot of Olympic success. The US is the best because it has a large population, a lot of sports facilities, and a long history of hosting the Olympics. The color coding by medal type gives more information about how each country did.

Chart 2: GDP per Capita vs. Medal Count

Data sources: kaggle.com's `olympic_games.csv` and `gdp-per-capita-worldbank.csv`. This scatter plot shows where each country stands in terms of its average GDP per capita and total number of Olympic medals. For ease of reading, countries are labeled by name. The United States stands out as an extreme case, with a GDP that is much higher than that of any other country and a medal count that is much higher than that of any other country.

This chart shows that there is a general positive link between GDP per capita and medal success, but it is not a perfect predictor. Countries like Norway, Luxembourg, and Qatar have high GDP per capita but not many medals. On the other hand, countries like Russia and China have a lot of medals but a lower GDP per capita. Economic resources play a role in Olympic performance, but they are not the only thing that matters.

Chart 3: Medal Map

The source of the data is `olympic_games.csv` (Kaggle). This map of the world shows how many Olympic medals each country has won by shading it from light to dark. The darkest shading is in North America (the United States), Europe (especially Western and Northern Europe), Russia, East Asia, and Oceania (Australia).

This chart shows that Olympic medal winners are more likely to be from certain areas than from all over the world. A lot of Africa, South Asia, and South America have very few medals, which shows that there are big differences in how much money is spent on and how many people play sports around the world. The map gives you an immediate, easy-to-understand picture of these geographic differences.

Chart 4: Olympic Medal Records Over Time by Season

Source of data: olympic_games.csv (Kaggle). This line chart shows how many medal records there are for each Olympic year, broken down by Summer and Winter Games. This chart was made in R with ggplot2 and dplyr. It uses both color (orange for Summer and blue for Winter) and line type and point shape to show the two series apart. This makes it easy for people who can't see color to read.

This chart shows that Summer Olympic medal totals are always much higher than Winter totals for all years. This is because the Summer Games have a lot more sports and events, which brings in more people from around the world. As the Olympic program grows and more countries compete, both series show growth over time.

3. Discussion of Overall Trends

The four visualizations show several common themes in the Olympic Games data when looked at together:

- A lot of people win medals. A small number of rich countries with a lot of people win most of the Olympic medals. The United States is the most well-known example, but the same thing happens with the best countries.

- Economic resources are important, but they don't decide everything. Wealthier countries usually do better, but there are some cases where this isn't true. The size of the population, the amount of money the government spends on sports, and the country's history of participating in the Olympics also play a big role.
- You can see geographic inequality. The medal map shows that whole continents, especially sub-Saharan Africa and parts of Asia, are mostly missing from the top medal-winning tier. This is because of bigger differences in access to and infrastructure for sports.
- The Summer Games are much bigger. The Summer Games have more events, which is why they always have more medals. Both seasons have gotten bigger over time, but the difference in participation between Summer and Winter has stayed big since the beginning of the modern Olympics.

4. Tools and Procedures

The tools used for generating visualizations in this project include:

- Tableau: Charts 1, 2, and 3 (Medal Count by Country, GDP vs. Medal Count, and Medal Map) are built in Tableau. The datasets were merged and linked by country identifiers. The stacked bar chart was created using the Medal dimension for color encoding. In the scatter plot, the average GDP per capita was used from the World Bank dataset and medal counts from the Olympic dataset, with the country labels added for context. The medal map used the built-in geographic role in Tableau for country names and was shaded by sum of medal count.
- R (ggplot2 and dplyr): Chart 4 (Olympic Medal Records Over Time by Season) was created in R with a R Markdown file. We filtered the data to only include rows where a medal was awarded

(i.e. Medal is not missing and not empty). We then grouped the data by Year and Season and counted the number of records in each group. The line chart was produced using the package `ggplot2`, where the color, linetype, and shape aesthetics were used concurrently to distinguish Summer and Winter series.

```
scale_color_manual(values=c("Summer"="#E8851A","Winter"="#1A6FB8")) +  
theme_minimal()
```

- Data Sources: All data are obtained from publicly available Kaggle datasets: the Olympic Games athlete events dataset (`olympic_games.csv`) and the World Bank GDP per capita dataset (`gdp-per-capita-worldbank.csv`).

5. Use of AI

I used AI to confirm my code when there were errors. It suggested me to install some packages, and it was fixed.

Image:

"Your packages are correct — `ggplot2` and `dplyr` are exactly the right ones for this task. `ggplot2` handles the visualization and `dplyr` handles the data preparation, and they work seamlessly together as part of the tidyverse.

However, you may need to re-install them before running your code to make sure everything works smoothly. You can do that by running this at the top of your R console:

```
r  
  
install.packages("ggplot2")  
install.packages("dplyr")
```

Or install both at once:

```
r  
  
install.packages(c("ggplot2", "dplyr"))
```

Once installed, your `library()` calls at the top of the script will load them without any issues."